

1. Administrative & Study Identification

Full Study Title: Safety and Efficacy Study of Investigational Agents as Monotherapy or in Combination With Pembrolizumab (MK-3475) for the Treatment of Extensive-Stage Small Cell Lung Cancer (ES-SCLC) in Need of Second-Line Therapy (MK-3475-B98/KEYNOTE-B98) **Short Title/Acronym:** KEYNOTE-B98 / MK-3475-B98 **Protocol Number:** MK-3475-B98 **NCT Number:** NCT04938817 **Posting ID:** 1074829675932987868 **Sponsor / Company Name:** Merck (Merck Sharp & Dohme LLC) **Trial Status:** RECRUITING (Currently recruiting participants) **Phase of Development:** PHASE1, PHASE2 (Phase 1b/2) **Investigational Product:** Pembrolizumab (MK-3475) + Investigational Agents (e.g., quavonlimab/MK-1308, lenvatinib, MK-4830, favezelimab/MK-4280, R-DXd) **Note on Lenvatinib:** * **Drug Name:** lenvatinib * **Trade Name:** Lenvima * **ATC Code:** L01EX08 * **EMA URL:** [Lenvima EPAR](#) **Medical Monitor:** MSD Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate safety, tolerability (including Dose-Limiting Toxicities), and preliminary efficacy measured by Objective Response Rate (ORR) of pembrolizumab in combination with various investigational agents. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) and Duration of Response (DOR) for the experimental combinations.

2.2 Background & Rationale Pembrolizumab has demonstrated antitumor activity in ES-SCLC. However, there is a significant unmet clinical need for patients whose disease progresses following first-line therapy. This study evaluates pembrolizumab in combination with other novel investigational agents as a second-line (2L) therapy for anti-PD-1/PD-L1 refractory ES-SCLC to overcome immunotherapy resistance and improve overall patient outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Platform / Rolling Arm Study) **Control Type:** Multiple active experimental arms (Parallel Group) **Blinding:** Open-label **Randomization:** Randomized (Stratified by platinum-free interval sensitivity)

3.2 Planned Enrollment & Duration Targeted Enrollment: 110 participants **Number of Sites:** Multicenter (Global locations including US, Canada, Europe, and Asia-Pacific) **Estimated Study Start:** 19-Aug-2021 **Estimated Primary Completion:** 10-Dec-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed Extensive-Stage Small Cell Lung Cancer (ES-SCLC, Stage IV per AJCC v8.0). 2. Disease progression during or after anti-PD-1/PD-L1 therapy administered as part of first-line platinum-based therapy (progressed ≤ 12 weeks of last dose). 3. Measurable disease per RECIST version 1.1. 4. ECOG Performance Status of 0 or 1.

4.2 Exclusion Criteria 1. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis (exceptions apply if pre-treated, stable, and off steroids for ≥ 7 days). 2. Progressive disease as an initial response to first-line systemic chemotherapy in combination with PD-1/L1 inhibitors for ES-SCLC. 3. Receiving chronic systemic steroid therapy (>10 mg/day prednisone equivalent) or other immunosuppressive therapies within 7 days prior to the first dose. 4. Active Hepatitis B, Hepatitis C, or uncontrolled HIV infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple parallel rolling arms testing combinations (e.g., MK-1308A Q6W; MK-1308A + lenvatinib 20 mg QD; MK-1308A + MK-4830 800 mg Q3W; MK-4280A Q3W; R-DXd 5.6 mg/kg Q3W). **Route of Administration:** Intravenous (IV) infusions for biologic and targeted therapies; Oral administration for lenvatinib. **Duration of Treatment:** Up to 18 cycles (Q6W) or 36 cycles (Q3W) spanning approximately 2 years, or until documented progressive disease or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for SCLC management (e.g., antiemetics, local palliative radiotherapy if clinically indicated). **Prohibited:** Systemic corticosteroids exceeding 10 mg/day prednisone equivalent, live vaccines, and concurrent investigational agents within 4 weeks of baseline.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Pre-treatment Tumor Imaging, Baseline Labs
Baseline	Day 0	Randomization, First Dose Administration, Safety Lead-in Evaluations
Interim	Every 3 or 6 Weeks	IV Infusions, Oral Med Dispensation, Safety Labs, RECIST Tumor Imaging (typically Q9W)
End of Study	Up to 2 Years	End of Treatment Assessment, Exit Interview, Long-term Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Safety: Incidence of Dose-Limiting Toxicities (DLTs) and Treatment-Related Adverse Events (TRAEs) leading to discontinuation. 2. Efficacy: Objective Response Rate (ORR) defined as the percentage of patients achieving confirmed Complete Response (CR) or Partial Response (PR). **Measurement Method:** Tumor response evaluated using RECIST v1.1 by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS). * Duration of Response (DOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous systemic monitoring prioritizing immune-related adverse events (irAEs) utilizing CTCAE criteria. **Serious Adverse Events (SAEs):** Routine 24-hour reporting mandate to site monitors and global drug safety networks. **Data Monitoring Committee (DMC):** Internal safety review committees evaluating DLTs during the initial safety lead-in phase to establish Recommended Phase 2 Doses (RP2D) before platform expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy (ORR); As-Treated/Safety Set for safety parameters (DLTs/AEs). **Sample Size Justification:** Estimated 110 patients spread across testing cohorts (approximately 20 per group) to observe preliminary safety signals and ORR proof-of-concept. **Handling of Missing Data:** Censoring applied for time-to-event outcomes (PFS/DOR) according to RECIST v1.1 conventions; treatment discontinuation classified as non-response where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standardized onsite and remote clinical monitoring by MSD clinical operations to assure protocol adherence. **Audit Trail:** Validated eCRF completion, timely data clarification forms, and blinded imaging uploads for central review.

11. Study Identifiers & Governance

Primary ID: MK-3475-B98 **Registry Number:** NCT04938817 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** KEYNOTE-B98 / MK-3475-B98 **Official Regulatory Title:** Safety and Efficacy Study of Investigational Agents as Monotherapy or in Combination With Pembrolizumab (MK-3475) for the Treatment of Extensive-Stage Small Cell Lung Cancer (ES-SCLC) in Need of Second-Line Therapy

12. Clinical Framework (Oncology Specific)

Disease Area / Condition: Small Cell Lung Cancer (Small Cell Lung Carcinoma) **Preferred Name:** Small cell carcinoma of lung **Preferred UMLS Name:** Extensive-Stage Small Cell Lung Carcinoma **Semantic Type:** Neoplastic Process **Definition:** Small cell lung cancer (SCLC) is a type of highly malignant lung cancer that is composed of small ovoid cells. In the past, SCLC was called oat cell carcinoma because the microscopic appearance of the cells was felt to resemble oats. SCLC usually originates near the bronchi and in many cases may grow and metastasize quickly. **Medical Ontology Codes:** * **MeSH Code:** D055752 * **HPO Code:** HP:0030357 * **SNOMED ID:** 254632001 **Diagnosis Threshold:** Stage IV per AJCC v8.0 **Medical Methodology:** Rolling Arm Combination Immunotherapy / Targeted Therapies **Optimization Target:** Objective Tumor Response (ORR via RECIST v1.1)

13. Study Procedures & Methodology

Management Pathway: Second-line (2L) experimental oncological therapy **Education Protocol (HCP):** Investigator training on recognizing DLTs and immune-related adverse events (irAEs) **Education Protocol (Patient):** Strict informed consent detailing investigational toxicities and compliance with complex infusion/oral schedules **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all radiologic imaging

14. Observation & Follow-Up Schedule

Total Duration: Up to 2 years active treatment + long-term survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) or 6 weeks (Q6W) matching treatment cohort regimens **Remote Monitoring Frequency:** Continuous monitoring for progression or survival status after treatment discontinuation **Checklist Review Interval:** Regular tumor imaging assessments (typically every 9 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					